From Promotions to Business Value

Your assignment

Meridian Rewards (fictional) is a multi-outlet retail business with a membership programme. It recently ran a multi-channel promotional exercise using discounts, buy-one-get-one (BOGO) offers and informational messages. Management now wants to improve the decisions it makes about future campaigns rather than simply commission another analysis.

Strategic objective: improve the contribution generated by member purchases over the next two quarters without increasing promotional expenditure, while retaining customer trust. An 8% improvement mentioned by the CEO is an aspiration, not evidence of achievable uplift.

Your group is the analytics advisory and delivery team. You must diagnose the decision problem, investigate the real data, assess competing analytics investments, build ONE functional information product for your selected near-term opportunity, and evaluate whether that product could generate meaningful value.

Exhibit 1 | Voices from the business

Stakeholder	Briefing
CEO	"We need profitable growth, not a fashionable AI model. Which information will actually change decisions?"
Marketing director	"We can adjust the future offer mix and test at most two simple targeting rules in a campaign. Give us a usable way to choose."
Finance director	"Reward redemptions and customer spend do not prove extra profit. Tell me what we know, what we assume, and how we could test the return."
Data manager	"We have the three source files, but permissions for individual targeting and the meaning of some event sequences have not been validated."
Operations manager	"Store-level labour forecasting is attractive, but these files have no store IDs, roster data or stock data."
Customer-experience lead	"Do not assume that a membership profile gives permission to use gender, age or income in targeting."

Exhibit 2 | The available source data

Use the THREE JSON-lines files supplied with this CAT (one JSON object per line). Identify the correct grain, keys and nested fields before calculating any metrics.

File / actual size	Contents	Critical handling point
portfolio.json / 10 offers	Offer ID, offer type, reward, difficulty, duration, advertised channels.	Channels are a LIST for each offer, not proof that any particular channel caused a response.
profile.json / 17,000 members	Member ID, age, gender, income, membership date.	Membership date is became_member_on. Missing income and age=118 occur together for 2,175 members.
transcript.json / 306,534 events	Member events: received, viewed, completed offers and transactions.	Received/viewed events use value["offer id"]; completed events use value["offer_id"]; transactions use value["amount"] and no offer ID.

Data boundary: Member transactions and promotion events are available, but there is no store identifier, member-level contact consent, product margin, verified offer-to-sale attribution, or randomized counterfactual. The time values are relative; their units and the comparability of future campaign periods must be stated as assumptions unless independently verified.

Exhibit 3 | Competing investment requests

Request	Set-up	Run / cycle	Effort	Earliest

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A. Campaign decision cockpit	£4,000	£500	2 weeks	3 weeks
B. Behavioural segment pilot workbench	£9,000	£1,000	5 weeks	8 weeks
C. Automated next-best-offer ML	£24,000	£2,000	10 weeks	16 weeks
D. Store labour-demand forecast	£7,000	£600	4 weeks	7 weeks

Portfolio constraints: £15,000 maximum initial expenditure; £1,500 maximum recurring expenditure per comparable campaign cycle; ONE analyst with eight person-weeks in a 10-week implementation window. Marketing can operate two distinct targeting rules. A has an accountable campaign manager; B has a marketing sponsor but unverified contact eligibility; C has no committed operational owner; D lacks the necessary store-level data.

Exhibit 4 | Illustrative business-case parameters

Assumption	Management planning value
Planning period	Four comparable future campaign cycles; treat one observed dataset period as a proxy ONLY for scenario illustration.
Currency comparability	Case costs are stated in £. For the classroom scenario, assume 1 dataset monetary unit = £1; the raw JSON does not itself specify currency.
A - possible incentive saving	Avoid 5% / 10% / 15% of recorded completed-offer reward face value per cycle (low/base/high); assume purchasing is unchanged. This is an unverified hypothesis.
B - possible incremental sales	Selected eligible target-cohort spending increases 1% / 3% / 5% per cycle (low/base/high); incremental-spend contribution rate = 35%, plus incremental incentive cost = 10% of incremental sales.
Economics	Count set-up once and running cost for all four cycles; no discounting needed. Exclude any claimed unmeasured effects from the base calculation.

Student tasks

1. Business needs and information requirements — 15 marks

1. Identify at least two underlying organisational needs (not just symptoms or proposed technologies). Justify the need you will address first using strategy and stakeholder evidence. [5]
2. Map two consequential decisions: decision owner, timing, available choices, information required, action triggered, and intended outcome. Show strategy → decision → information → action → value. [7]
3. Specify two leading and two lagging indicators, noting an important failure or gaming risk in one metric. [3]

2. Data investigation and fitness for decision — 15 marks

4. Document file grain, keys, event-specific fields, join logic, filtering decisions and at least two reproducible data-quality checks. [5]
5. Compute at least four defensible metrics and compare at least two offer types or behavioural cohorts. Include member spending, purchasing members and offer-event measures with explicit denominators. [6]
6. Explain important missingness, event-sequence and attribution limitations; label observed fact, inference and assumption. [4]

3. Analytics portfolio and prioritisation — 20 marks

7. Frame three or more proposals as decision-focused use cases, including named user, changed decision, information product, operational action and value pathway. [5]
8. Apply hard gates before ranking: accountable owner, lawful/usable data, acceptable risks and an operational ability to act. Specify prerequisites when a gate fails. [5]
9. Score eligible opportunities on 1–5 value (V), feasibility (F), actionability (A) and residual risk (R) with evidence-based justification. Calculate $P = 0.35V + 0.25F + 0.20A + 0.20(6 - R)$. Higher R means greater remaining risk. [6]
10. Select ONE near-term initiative for the actual product build, sequence or defer the others, and show feasibility under all constraints. Justify one rejected alternative. [4]

4. Design and BUILD a working information product — 25 marks

Your primary final output is an operational prototype tied to the use case you selected—not a static set of graphs. It may be built in Power BI, Excel, Streamlit, a notebook with functional input controls, or a comparable accessible tool. A machine-learning model is NOT required.

11. Implement reproducible calculations from the supplied files plus useful interactive controls (e.g., offer type, period, member behaviour cohort, scenario) and at least three interpretable decision KPIs. [14]
12. Produce an explicit action-oriented output: e.g., a campaign decision summary, recommended offer-mix comparison, scenario comparison or a privacy-safe cohort brief. Describe who will use it and when. [6]
13. Demonstrate operation on the data supplied: correct denominators, readable definitions, warning/quality indicators, reproducible refresh steps, documented limitations and a successful live walkthrough. [5]

5. Responsible deployment and defence — 5 marks

14. Identify two product-specific privacy/fairness/data-use risks and enforceable safeguards. [3]
15. Present a feasible 10-week go/no-go sequence and defend the decision and working product as a group. [2]

What to submit

- A concise executive decision brief: up to 6 pages excluding figures, references and calculation annex. Start with the investment recommendation, not a methodology narrative.
- Reproducible analysis files and a data dictionary/metric sheet; include the prioritisation matrix and editable business-value scenario calculations.
- A maximum 8-slide executive presentation; 10-minute group presentation plus questions. Each member must be able to explain product logic and value limitations.

Assessment principle: The product is judged by fitness to a meaningful decision, analytical validity, usability, actionability and credible value verification—not graphical polish, algorithmic sophistication or the size of claimed ROI.